

KRALJEVO, Serbia

WMO: 133760

Lat: 43.700N

Lon: 20.700E

Elev: 712

StdP: 14.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	17.4	7.5	8.3	14.4	12.2	10.5	18.1	20.2	42.4	17.7	37.2	2.7	320	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	93.8	69.7	90.3	69.6	87.2	68.9	73.6	86.3	71.8	84.8	70.2	83.0	5.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	113.1	79.0	67.9	105.3	76.6	66.1	98.9	75.4	37.6	85.7	36.0	84.2	34.6	83.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	13.5	11.4	DB	2.6	98.3	8.6	4.0	-3.5	101.2	-8.5	103.5	-13.3	105.8	-19.5	108.7
			WB	2.5	76.3	8.3	2.1	-3.4	77.9	-8.2	79.1	-12.9	80.3	-18.9	81.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	33.0	37.2	44.6	53.7	61.7	68.6	72.0	71.9	63.2	53.7	45.0	35.1	(6)
	DBStd	15.49	8.80	9.12	8.14	6.74	6.22	6.10	5.39	5.29	6.53	7.31	8.13	8.04	(7)
	HDD50	1851	530	364	205	39	2	0	0	0	1	48	195	467	(8)
	HDD65	4924	993	777	633	340	139	34	8	8	108	355	602	927	(9)
	CDD50	3085	2	7	37	151	363	557	681	679	398	162	43	5	(10)
	CDD65	683	0	0	0	2	35	140	223	223	55	4	1	0	(11)
	CDH74	7779	0	0	3	93	506	1544	2466	2403	682	80	2	0	(12)
	CDH80	3131	0	0	0	17	106	582	1086	1094	237	9	0	0	(13)
(14) Wind	WSAvg	3.8	3.8	4.3	4.7	4.5	3.6	3.5	3.3	3.2	3.3	3.4	4.1	3.7	(14)
(15) Precipitation	PrecAvg	31.0	1.9	1.9	2.5	2.7	3.4	3.4	2.8	2.4	2.8	2.5	2.2	2.4	(15)
	PrecMax	46.3	4.3	3.0	6.2	7.0	7.7	7.6	7.6	5.6	6.8	5.5	4.4	4.8	(16)
	PrecMin	21.2	0.4	0.6	0.4	1.1	1.5	1.1	0.6	0.0	0.5	0.1	0.1	0.3	(17)
	PrecStd	5.5	1.1	0.7	1.4	1.3	1.5	1.7	1.7	1.5	1.6	1.4	1.1	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	65.9	73.3	82.3	86.4	93.4	99.2	100.2	92.8	80.6	71.5	62.3	(19)
		MCWB	49.8	52.0	54.8	61.4	66.2	71.2	69.6	69.3	67.3	63.0	58.0	52.7	(20)
	2%	DB	53.4	60.6	68.8	75.8	82.7	89.7	93.7	94.6	86.3	75.6	66.6	56.0	(21)
		MCWB	46.4	50.0	52.9	58.7	65.1	70.8	70.2	69.3	65.8	61.7	56.3	48.8	(22)
	5%	DB	49.6	55.7	64.5	71.7	80.1	86.4	89.8	90.1	81.0	71.6	62.6	51.0	(23)
		MCWB	44.4	47.2	50.9	56.2	64.2	69.6	70.2	68.9	64.4	60.2	54.0	46.4	(24)
	10%	DB	45.3	51.5	60.0	67.9	76.3	82.9	86.2	86.2	76.9	67.5	58.7	46.4	(25)
		MCWB	41.7	45.1	49.1	54.8	62.6	68.0	69.5	68.6	63.3	58.6	52.1	43.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	53.5	56.3	63.2	69.0	76.1	76.3	75.5	70.1	65.6	59.6	53.3	(27)
		MCDB	57.9	63.1	69.8	77.9	81.2	86.9	85.1	87.4	82.9	77.3	68.5	60.8	(28)
	2%	WB	47.3	50.5	53.8	59.9	66.9	73.0	73.6	72.9	67.7	63.0	56.9	49.6	(29)
		MCDB	52.6	59.4	65.7	73.0	79.2	85.3	86.4	86.2	80.8	72.0	64.7	54.9	(30)
	5%	WB	44.6	48.1	51.8	57.8	65.3	70.8	71.8	71.0	66.1	61.1	54.7	46.5	(31)
		MCDB	48.8	54.6	62.7	69.3	77.0	83.0	85.0	84.1	78.2	69.3	61.4	50.8	(32)
	10%	WB	41.9	45.6	49.7	55.7	63.5	69.1	70.1	69.5	64.4	59.1	52.5	43.4	(33)
		MCDB	45.1	51.1	59.4	65.2	74.2	80.9	83.2	82.3	74.6	66.9	58.2	46.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	16.2	19.2	21.0	21.7	21.8	23.4	24.4	21.8	20.5	16.9	12.6	(35)
		MCDBR	20.5	26.2	31.0	30.7	29.4	28.9	31.4	33.4	31.4	28.7	23.9	20.4	(36)
	5% WB	MCWBR	14.1	16.2	16.8	15.2	13.5	12.1	11.5	11.7	13.1	15.5	14.4	14.4	(37)
		MCWBR	19.5	23.6	27.3	26.3	25.9	26.0	26.6	26.4	26.5	25.1	21.7	18.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.336	0.364	0.403	0.463	0.455	0.446	0.447	0.459	0.436	0.394	0.354	0.333	(40)
		taud	2.341	2.239	2.193	2.054	2.137	2.208	2.206	2.169	2.205	2.315	2.372	2.366	(41)
	Ebn at Noon		246	258	262	255	261	263	261	253	249	245	237	235	(42)
		Edh at Noon	26	34	41	50	48	45	44	44	40	31	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		441	701	1059	1378	1683	1904	1960	1773	1240	862	545	383	(44)
	RadStd		65	85	131	165	171	189	154	167	127	109	61	60	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.81	N/A	N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+1.17	N/A	N/A	N/A	+0.70	N/A	N/A	-268	+189	N/A

Nomenclature: See separate page